When stars exhaust their nuclear fuel, the outward pressure from nuclear fusion in the star's core diminishes and the star begins to implode under its own weight. As the star's core crashes in on itself, its temperature rapidly rises, sometimes resulting in an enormous explosion that blows off the star's outer layers in a brilliant display of heavenly fireworks. Such an explosion is known as a supernova; for a period of weeks, a single exploding star can burn as bright as a billion suns. It's truly mind-boggling: a single star burning as bright as almost an entire galaxy! Different types of stars— of different sizes, with different atomic abundances, and so on—give rise to different kinds of supernova explosions, but for many years astronomers have realized that certain supernova explosions always seem to burn with the same intrinsic brightness. These are *type Ia* supernova explosions.

In a type Ia supernova, a white dwarf star—a star that has exhausted its supply of nuclear fuel but has insufficient mass to ignite a supernova explosion on its own—sucks the surface material from a nearby companion star. When the dwarf star's mass reaches a particular critical value, about 1.4 times that of the sun, it undergoes a runaway nuclear reaction that causes the star to go supernova. Since such supernova explosions occur when the dwarf star reaches the same critical mass, the characteristics of the explosion, including its overall intrinsic brightness, are largely the same from episode to episode. Moreover, since supernovae, unlike 100-watt lightbulbs, are so fantastically powerful, not only do they have a standard, dependable brightness but you can also see them clear across the universe. They are thus prime candidates for standard candles.[22]

In the 1990s, two groups of astronomers, one led by Saul Perlmutter at the Lawrence Berkeley National Laboratory, and the other led by Brian Schmidt at the Australian National University, set out to determine the deceleration—and hence the total mass/energy—of the universe by measuring the recession speeds of type Ia supernovae. Identifying a supernova as being of type Ia is fairly straightforward because the light their explosions generate follows a distinctive pattern of steeply rising then gradually falling intensity. But actually catching a type Ia supernova in the act is no small feat, since they happen only about once every few hundred years in a typical galaxy. Nevertheless, through the innovative technique of simultaneously observing thousands of galaxies with wide-